

One Mark Questions

01. Riser is designed so as to (GATE-ME-87)
 (a) Freeze after the casting freezes
 (b) Freeze before the casting freezes
 (c) Freeze at the same as the casting
 (d) Minimize the time of pouring
02. The contraction allowance provided on the pattern and core boxes compensates for the following type of contraction. (GATE-ME-88)
 (a) Liquid contraction
 (b) Solidification contraction
 (c) Solid contraction
 (d) All the above three types of contractions.
03. Chills are used in molds to (GATE-ME-89)
 (a) achieve directional solidification
 (b) reduce possibility of blow holes
 (c) reduce the freezing time
 (d) smoothen the metal for reducing spatter.
04. Increase in water content in molding sand causes (GATE-ME-89)
 (a) flowability to go through a maxima
 (b) permeability to go through a maxima
 (c) compressive strength to go through a maxima
 (d) strength to go through a maxima
05. The pressure at the in-gate will be maximum with the gating system (GATE-ME-90)
 (a) 4:8:3 (b) 1:3:3 (c) 1:2:4 (d) 1:2:1
06. When there is no room temperature change, the *total shrinkage allowance* on a pattern is INDEPENDENT OF (GATE-ME-91)
 (a) Pouring temperature of the liquid metal
 (b) Freezing temperature of the liquid metal
 (c) The component size
 (d) Coefficient of thermal contraction of solidified metal
07. Converging passage is used for feeding the liquid molten metal into the mould to (GATE-ME-91)
 (a) Increase the rate of feeding
 (b) Quickly break off the protruding portion of the casting
 (c) Decrease wastage of cast metal
 (d) Avoid aspiration of air
08. In a green sand moulding process, uniform ramming leads to (GATE-ME-92)
 (a) Less chance of gas porosity
 (b) Uniform flow of molten metal into the mold cavity
 (c) Greater dimensional stability of the casting
 (d) Less sand expansion type of casting defect
09. Centrifugally casted products have (GATE-ME-93)
 (a) Large grain structure with high porosity
 (b) Fine grain structure with high density
 (c) Fine grain structure with low porosity
 (d) Segregation of slug towards the outer skin of the casting
10. Light impurities in the molten metal are prevented from reaching the mold cavity by providing (GATE-ME-96)
 (a) Strainer (b) Bottom well
 (c) Skim bob (d) All of the above
11. Which of the following materials requires the largest shrinkage allowance, while making a pattern for casting? (GATE-ME-99)
 (a) Aluminum (b) Brass
 (c) Cast iron (d) Plain carbon steel

12. Disposable patterns are made of (GATE-ME-2000)
 (a) wood (b) rubber
 (c) metal (d) polystyrene
13. Shrinkage allowance on pattern is provided to compensate for shrinkage when (GATE-ME-01)
 (a) the temperature of liquid metal drops from pouring to freezing temperature
 (b) the metal changes from liquid to solid state at freezing temperature
 (c) the temperature of solid phase drops from freezing to room temperature
 (d) the temperature of metal drops from pouring to room temperature
14. In centrifugal casting, the lighter impurities are (GATE-ME-02)
 (a) uniformly distributed
 (b) forced towards the outer surface
 (c) trapped near the mean radius of the casting
 (d) collected at the center of the casting
15. The primary purpose of sprue in casting mold is to (GATE-ME-02)
 (a) Feed the casting at rate consistent with the rate of solidification
 (b) Act as a reservoir for molten metal
 (c) Feed molten metal from the pouring basin to the gate.
 (d) Help feed the casting until all solidification takes place
16. Hardness of green sand mould increases with (GATE-ME-03)
 (a) increase in moisture content beyond 6 percent
 (b) increase in permeability
 (c) decrease in permeability
 (d) increase in both moisture content and permeability
17. Misrun is a casting defect which occurs due to (GATE-ME-04)
 (a) very high pouring temperature of the metal
 (b) insufficient fluidity of the molten metal
 (c) absorption of gases by the liquid metal
 (d) improper alignment of the mould flasks
18. An expendable pattern is used in (GATE-ME-06)
 (a) slush casting (b) squeeze casting
 (c) centrifugal casting (d) investment casting
19. Which of the following engineering materials is the most suitable candidate for hot chamber die casting? (GATE-ME-07)
 (a) low carbon steel (b) titanium
 (c) copper (d) tin
20. Two streams of liquid metal which are not hot enough to fuse properly result into a casting defect known as (GATE-ME-09)
 (a) cold shut (b) swell
 (c) blow hole (d) scar
21. Green sand mould indicates that (GATE-ME-11)
 (a) polymeric mould has been cured
 (b) mould has been totally dried
 (c) mould is green in colour
 (d) mould contains moisture
22. Negative allowance is provided on the pattern to take care of (GATE-PI-89)
 (a) The distortion allowance
 (b) The draft allowance
 (c) The machining allowance
 (d) The shake allowance
23. The process by which shell like castings (like toys) are made is (GATE-PI-89)
 (a) shell molding (b) centrifugal casting
 (c) slush casting (d) die casting
24. Poor collapsibility of the mold leads to type of casting defect in aluminum castings. (GATE-PI-89)

25. For casting of turbine blades made of high temperature and high strength alloys, the most suitable process is **(GATE-PI-90)**
 (a) Die casting (b) Investment casting
 (c) Centrifugal casting (d) Slush casting
26. For the same material, powder metallurgy process is superior to casting for **(GATE-PI-89)**
 (a) making large products
 (b) better control over the density of product
 (c) better strength of the finished product
 (d) making parts with wide variations of thickness at different sections
27. Green strength in powder metallurgy refers to the strength of **(GATE-PI-89)**
 (a) original material in the bulk form
 (b) the powder before compaction
 (c) the powder after compaction
 (d) the powder after sintering and cooling
28. The problem of shrinkage cavity is more likely to occur while casting....A..... B..... a chill block **(GATE-PI-90)**
 A. metal / plastic B. with / without
29. **Assertion:** Converging passage is used for feeding liquid metal into a mould.
Reason: Inhalation of air promotes blow holes in casting. **(GATE-PI-90)**
30. The optimum pouring time for a casting depends on several factors. One important factor among them is **(GATE-PI-91)**
 (a) location of riser
 (b) porosity of sand mold
 (c) fluidity of casting metal
 (d) area of the pouring basin
31. Ceramic slurry is used in making the molds for **(GATE-PI-91)**
 (a) investment casting (b) die casting
 (c) shell mold casting (d) centrifugal casting
32. Low permeability molds and low pouring temperature of the molten metal causes defect in castings. **(GATE-PI-91)**
33. Pressure die casting deals with casting metals of relativelyA..... melting point and providesB..... production rate. **(GATE-PI-91)**
 A. higher / lower B. higher / lower
34. Increase in carbon content in plain carbon steels raise its **(GATE-PI-92)**
 (a) Ductility and UTS
 (b) Tensile strength and malleability
 (c) Tensile strength and hardness
 (d) Ductility and melting temperature
35. Investment casting uses ... A..... as pattern material and is preferred for small parts withB designs. **(GATE-PI-92)**
 A. wax / resin coated sand B. simple / intricate
36. The primary function of a riser is to **(GATE-PI-92)**
 (a) feed molten metal to casting as it solidifies
 (b) prevent atmospheric air from contaminating the metal in the mold
 (c) allow gases to easily escape from mold cavity
 (d) allow molten metal to rise above the mold cavity
37. Chaplets are placed between mold in order to **(GATE-PI-92)**
 (a) promote directional solidification
 (b) help alloying the metal
 (c) facilitate easy removal of core from casting
 (d) prevent core movement due to buoyancy
38. Investment casting uses (wax/resin coated sand) as pattern material and is preferred for small parts with..... (simple/intricate) designs. **(GATE-PI-92)**
39. Bentonite is commonly used bonding material for dry sand cores. (T/F) **(GATE-PI-94)**

Shell molding is an investment casting process.
(T/F) (GATE-PI-95)

41. Which one of the following is NOT a property of a sand mold? (GATE-PI-03)
(a) Permeability (b) Collapsibility
(c) Strength (d) Fluidity

42. Wood flour is added to core sand to improve: (GATE-PI-04)
(a) Collapsibility of core (b) Dry strength of core
(c) Shear strength of core (d) Tolerance on casting

43. An expendable pattern is used in (GATE-PI-05)
(a) slush casting (b) squeeze casting
(c) centrifugal casting (d) investment casting

44. Cold shut is a defect in casting due to (GATE-PI-06)
(a) sand sliding from the cope surface
(b) internal voids or surface depression due to excessive gas trapped
(c) discontinuity resulting from hindered contraction
(d) two streams of material that are too cold to fuse properly

45. In hollow cylindrical parts made by centrifugal casting, the density of the part is (GATE-PI-08)
(a) maximum at the outer region
(b) maximum at the inner region
(c) maximum at the mid point between outer and inner surfaces
(d) uniform throughout

46. Hot chamber die casting is NOT suited for (GATE-PI-09)
(a) Lead and its alloys
(b) Zinc and its alloys
(c) Tin and its alloys
(d) Aluminum and its alloys

47. Solidification time of a metallic alloy casting is (GATE-PI-10)

- (a) Directly proportional to its surface area
- (b) Inversely proportional to the specific heat of the cast material
- (c) Inversely proportional to the *thermal diffusivity* of the mould material
- (d) Inversely proportional to the pouring temperature

48. In sand casting fluidity of the molten metal increases with (GATE-PI-11)

- (a) Increase in degree of superheat
- (b) Decrease in pouring rate
- (c) Increase in thermal conductivity of the mold
- (d) Increase in sand grain size.

49. Which of the following casting processes uses expendable pattern and expendable mold? (GATE-PI-11)

- (a) Shell mould casting (b) Investment casting
- (c) Pressure die casting (d) Centrifugal casting

50. A cube shaped casting solidifies in 5 min. The solidification time in min for a cube of the same material, which is 8 times heavier than the original casting, will be (GATE-ME & PI-13)
(a) 10 (b) 20 (c) 24 (d) 40

51. Match the casting defects (Group A) with the probable causes (Group B): (GATE-ME-14-SET-3)

Group A	Group B
P: Hot tears	1: Improper fusion of two streams of liquid metal
Q: Shrinkage	2: Low permeability of the sand mould
R: Blow holes	3: Volumetric contraction both in liquid and solid stage
S: Cold shut	4: Differential cooling rate

- (a) P-1, Q-3, R-2, S-4 (b) P-4, Q-3, R-2, S-1
- (c) P-3, Q-4, R-2, S-1 (d) P-1, Q-2, R-4, S-3

52. The hot tearing in a metal casting is due to
 (a) high fluidity (GATE-ME-14-SET-3)
 (b) high melt temperature
 (c) wide range of solidification temperature
 (d) low coefficient of thermal expansion

53. Chills are used in casting moulds to (GATE-PI-14)
 (a) achieve directional solidification
 (b) reduce the roughness of top surface of the cast product
 (c) increase the solidification time
 (d) reserve excess molten metal

54. In full mould (cavity-less) casting process, Pattern is made of (GATE -15 -Set 3)
 (a) Expanded polystyrene (b) Wax
 (c) Epoxy (d) Plaster of Paris

55. Match the items in the first column to their functions in the second column.

P. Sprue	1. Regulates flow of molten metal into mould cavity
Q. Riser	2. Feeds molten metal from pouring basin to gate
R. Gate	3. Acts as a reservoir for molten metal
S. Pouring basin	4. Supplies molten metal to compensate for liquid shrinkage

(GATE - PI -15)

- (a) P-1,Q-2,R-3,S-4 (b) P-2,Q-4,R-1,S-3
 (c) P-4,Q-2,R-1,S-3 (d) P-2,Q-4,R-3,S-1

56. The part of a gating system which regulates the rate of pouring of molten metal is (GATE-16-SET-1)
 (a) pouring basin (b) runner
 (c) choke (d) ingate

57. The welding process which uses a blanket of fusible granular flux is (GATE - 16 - SET - 2)
 (a) tungsten inert gas welding
 (b) submerged arc welding
 (c) electroslag welding
 (d) thermit welding

58. Equal amounts of a liquid metal at the same temperature are poured into three moulds made of steel, copper and aluminum. The shape of the cavity is a cylinder with 15 mm diameter. The size of the moulds are such that the outside temperature of the moulds do not increase appreciably beyond the atmospheric temperature during solidification. The sequence of solidification in the mould from the fastest to slowest is
 (Thermal conductivities of steel, copper and aluminum are 60.5, 401 and 237 W/m-K, respectively. Specific heats of steel, copper and aluminum are 434, 385 and 903 J/kg-K, respectively. Densities of steel, copper and aluminum are 7854, 8933 and 2700 kg/m³, respectively.)

(GATE-16-SET-3)

- (a) Copper - Steel - Aluminum
 (b) Aluminum - Steel - Copper
 (c) Copper - Aluminum - Steel
 (d) Steel - Copper - Aluminum

59. In carbon dioxide molding process, the binder used is (GATE - PI - 17)

- (a) Sodium bentonite
 (b) Calcium bentonite
 (c) Sodium silicate
 (d) Phenol formaldehyde

60. During solidification of a pure molten metal, the grains in the casting near the mould wall are (GATE - 18 - SET - 2)

- (a) coarse and randomly oriented
 (b) fine and randomly oriented
 (c) fine and ordered
 (d) coarse and ordered

61. In a casting process, a vertical channel through which molten metal flows downward from pouring basin to runner for reaching the mold cavity is called (GATE-19-SET-1)

- (a) riser (b) sprue
 (c) blister (d) pin hole

62. The fluidity of molten metal of cast alloys (without any addition of fluxes) increases with increase in
(GATE - 19 - SET - 2)

- (a) surface tension
(b) degree of superheat
(c) viscosity
(d) freezing range

63. Match the component with the corresponding manufacturing process in the table below.

Component		Manufacturing process
P Aluminum alloy piston for IC engine	1	Blow molding
Q Low carbon steel oil pan	2	Powder metallurgy
R Tungsten carbide cutting tool insert	3	Sand casting
S Plastic bottle	4	Deep drawing

(GATE-PI-21)

- (a) P-1, Q-3, R-2, S-4
(b) P-3, Q-2, R-1, S-4
(c) P-3, Q-4, R-2, S-1
(d) P-2, Q-3, R-4, S-1

64. Consider sand casting of a cube of edge length a . A cylindrical riser is placed at the top of the casting. Assume solidification time, $t_s \propto V/A$, where V is the volume and A is the total surface area dissipating heat. If the top of the riser is insulated, which of the following radius/radii of riser is/are acceptable?

(GATE-22-SET-2)

- (a) $\frac{a}{3}$ (b) $\frac{a}{2}$ (c) $\frac{a}{4}$ (d) $\frac{a}{6}$

65. Match the casting methods with products.

Casting method	Products
P Continuous casting	1 Thin and intricate shaped components
Q Investment casting	2 Hollow axisymmetric parts (such as pipes)
R Centrifugal casting	3 Slabs and strips

(GATE-PI-22)

- (a) P-3, Q-1, R-2
(b) P-2, Q-3, R-1
(c) P-3, Q-2, R-1
(d) P-2, Q-1, R-3

66. In a metal casting process to manufacture parts, both patterns and moulds provide shape by dictating where the material should or should not go. Which of the option(s) given correctly describe(s) the mould and the pattern? (GATE_ME-23)

- (a) Mould walls indicate boundaries within which the molten part material is allowed, while pattern walls indicate boundaries of regions where mould material is not allowed.
(b) Moulds can be used to make patterns.
(c) Pattern walls indicate boundaries within which the molten part material is allowed, while mould walls indicate boundaries of regions where mould material is not allowed.
(d) Patterns can be used to make moulds.

Two Marks Questions

01. Two cubical castings of the same metal and sizes of 2 cm side and 4 cm side are moulded in green sand. If the smaller casting solidifies in 2 mins, the expected time of solidifications of larger casting will be (GATE-ME-89)

- (a) 16 min (b) $2\sqrt{8}$ min
(c) 8 min (d) 4 min

02. Match the following: (ME-GATE-92)

Casting Process	Product
A. Slush casting	1. Turbine blade
B. Shell molding	2. Machine tool bed
C. Dry sand molding	3. Cylindrical block
D. Centrifugal casting	4. Hollow castings like lamp shades
	5. Rain water pipes
	6. Cast iron shoe brake

03. Only 4 pairs can be matched (GATE-ME-94)

Casting Process	Product
A. Investment casting	1. Turbine rotors
B. Die casting	2. Turbine blades
C. Centrifugal casting	3. Connecting rods
D. Drop forging	4. Galvanized iron pipes
E. Extrusion	5. Cast iron pipes
F. Shell molding	6. Carburetor body

04. Match the following (GATE-ME-98)

Group – I

- A. Sand casting
- B. Plaster mold casting
- C. Shell mold casting
- D. Investment casting

Group – II

- 1. Symmetrical and circular shapes only
- 2. Parts have hardened skins and soft interior
- 3. Minimum post casting process
- 4. Parts have soft skin and hard interior
- 5. Parts have tendency to warp
- 6. Suitable only for non-ferrous metals.

 05. With a solidification factor of $0.97 \times 10^6 \text{ s/m}^2$, the solidification time (in seconds) for a spherical casting of 200 mm diameter is (GATE-ME-03)

- (a) 539
- (b) 1078
- (c) 4311
- (d) None

 06. Gray cast iron blocks $200 \times 100 \times 10 \text{ mm}$ are to be cast in sand moulds. Shrinkage allowance for pattern making is 1%. The ratio of the volume of pattern to that of the casting will be

(GATE-ME-04)

- (a) 0.97
- (b) 0.99
- (c) 1.01
- (d) 1.03

07. Match List – I (products) with List – II (casting process) and select the correct answer using the codes given below the lists (GATE-ME-05)

List – I

- A. Hollow statues
- B. Dentures
- C. Aluminum alloy pistons
- D. Rocker arm

List – II

- 1. Centrifugal casting
- 2. Investment casting
- 3. Slush casting
- 4. Shell moulding
- 5. Gravity die casting

- (a) A-3, B-2, C-4, D-5
- (b) A-1, B-3, C-4, D-5
- (c) A-1, B-2, C-3, D-4
- (d) A-3, B-2, C-5, D-4

 08. A mould has a down sprue whose length is 20 cm and the cross sectional area at the base of the down sprue is 1 cm^2 . The down sprue feeds a horizontal runner leading into the mould cavity of volume 1000 cm^3 . The time required to fill the mould cavity will be (GATE-ME-05)

- (a) 4.05 s
- (b) 5.05 s
- (c) 6.05 s
- (d) 7.25 s

 09. In a sand casting operation, the total liquid head is maintained constant such that it is equal to the mould height. The time taken to fill the mold with a top gate is t_A . If the same mold is filled with a bottom gate, then the time taken is t_B . Ignore the time required to fill runner and frictional effects. Assume atmospheric pressure at the top molten metal surfaces. The relation between the t_A and t_B is (GATE-ME-06)

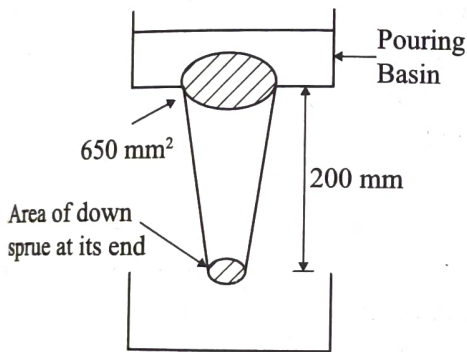
- (a) $t_B = \sqrt{2} \cdot t_A$
- (b) $t_B = 2 \cdot t_A$
- (c) $t_B = \frac{t_A}{\sqrt{2}}$
- (d) $t_B = 2\sqrt{2} \cdot t_A$

10. If a particular Fe-C alloy contains less than 0.83% carbon, it is called (GATE-ME-07)

- (a) High speed steel
- (b) Hypo-eutectoid steel
- (c) Hyper eutectoid steel
- (d) Cast irons

11. Volume of a cube of a side " l " and volume of sphere of radius ' r ' are equal. Both the cube and the sphere are solid and same material. They are being cast. The ratio of the solidification time of the cube to the same of the sphere is (GATE-ME-07)
- (a) $(4\pi/6)^3 \cdot (r/l)^6$ (b) $(4\pi/6) \cdot (r/l)^2$
 (c) $(4\pi/6)^2 \cdot (r/l)^3$ (d) $(4\pi/6)^2 \cdot (r/l)^4$

12. A 200 mm long down sprue has an area of cross section of 650 mm² where the pouring basin meets the down sprue (i.e., at the beginning of the down sprue). A constant head of molten metal is maintaining by the pouring basin. The molten metal flow rate is 6.5×10^5 mm³/s. Considering the end of down sprue to be open to atmosphere and an acceleration due to gravity of 10^4 mm/s², the area of the down sprue in mm² at its end (avoiding aspiration effect) should be (GATE-ME-07)



- (a) 650 (b) 350 (c) 290.7 (d) 190.0
13. While cooling, a cubical casting of side 40 mm undergoes 3%, 4% and 5% volume shrinkage during the liquid state, phase transition and solid state respectively. The volume of metal compensated from the riser is (GATE-ME-08)
- (a) 2% (b) 7% (c) 8% (d) 9%

14. Match the following (GATE-ME-09)

Part used in casting	Purpose
P. Metallic chills	1. Support for the core
Q. Metallic chaplets	2. Reservoir of the molten metal

R. Riser	3. Control cooling of critical sections
S. Exothermic padding	4. Progressive solidification

- (a) P-1, Q-3, R-2, S-4 (b) P-1, Q-4, R-2, S-3
 (c) P-3, Q-4, R-2, S-1 (d) P-4, Q-1, R-2, S-3
15. In a gating system, the ratio 1:2:4 represents (GATE-ME-10)
- (a) Sprue base area: Runner area: In-gate area
 (b) Pouring basin area: In-gate area: Runner area
 (c) Sprue base area: In-gate area: Casting area
 (d) Runner area: In-gate area: Casting area
16. For the case of molding, it is decided to replace a spherical riser of diameter 100 mm by a cylindrical riser. Determine size of the cylindrical riser that will have the identical solidification time. Assume blind risering & ratio of height to radius of cylinder is 2. (GATE-ME-10)
- (a) 10 cm (b) 100 cm
 (c) 50 cm (d) 5 cm
17. A cubic casting of 50 mm side undergoes volumetric solidification shrinkage and volumetric solid contraction of 4% and 6% respectively. No riser is used. Assume uniform cooling in all directions. The side of cube after solidification and contraction is (GATE-ME-11)
- (a) 48.32 mm (b) 49.90 mm
 (c) 49.94 mm (d) 49.96 mm
18. Shrinkage allowance on pattern is provided to compensate for shrinkage when (GATE-PI-89)
- (a) the temperature of liquid metal drops from pouring to freezing temperature
 (b) the metal changes from liquid to solid state at freezing temperature
 (c) the temperature of solid phase drops from freezing to room temperature
 (d) the temperature of metal drops from pouring to room temperature

19. What is the velocity of the steel at the bottom of the sprue if the sprue height is 30 cm? assume the frictional and other losses to be 20%. (GATE-PI-89)
20. When there is no room temperature change, the total shrinkage allowance on a pattern is INDEPENDENT OF (GATE-PI-90)
- pouring temperature of the liquid metal
 - freezing temperature of the liquid metal
 - the component size
 - coefficient of thermal contraction of solidified metal
21. Match the application with type of pattern (GATE-PI-92)
- Application**
- Undercut in components
 - Large bells
 - Mass production of casting by Machine molding
 - Components with irregular parting lines
- Type of pattern**
- Cope and drag
 - Follow board
 - Gated
 - Loose piece
 - Sweep
22. Match the cast defect with their cause (GATE-PI-93)
- | List - I (Defect) | List - II (Cause) |
|---------------------|-----------------------------|
| A. Hot tears | 1. Mould restraint |
| B. Porosity | 2. Inadequate risering |
| C. Sand inclusions | 3. High pouring temperature |
| D. Shrinkage cavity | 4. Loosely rammed sand mold |
| | 5. Gas entrapment |
23. According to Chvorinov's rule, the solidification time of a casting is proportional to $[\text{Volume/Surface area}]^n$ where $n = \dots\dots\dots$ (GATE-PI-94)
- 0.5
 - 1
 - 2
 - 4
24. Match the cast defect with their cause (GATE-PI-95)
- Casting Defect**
- Hot tears
 - Porosity
 - Sand inclusions
 - Shrinkage cavity
- Most Appropriate Cause**
- Mold restraint
 - Inadequate risering
 - High pouring temperature
 - Loosely rammed sand mold
 - Gas entrapment
25. The height of the down sprue is 175 mm and its CS area at the base is 200 mm². The CS area of the horizontal runner is also 200 mm². Assuming no losses, indicate the correct choice for the time (seconds) required to fill a mold cavity of volume 10⁶ mm³. (use $g = 10 \text{ m/s}^2$) (GATE-PI-02)
- 2.67
 - 8.45
 - 26.72
 - 84.50
26. The permeability of molding sand was determined using a standard AFS sample by passing 2000 cc of air at a gauge pressure of 10 g/cm². If the time taken for the air to escape was 1 min, the permeability number is: (GATE-PI-02)
- 112.4
 - 100.2
 - 75.3
 - 50.1
27. Proper gating design in metal casting (GATE-PI-02)
- Influences the freezing range of the melt
 - Compensates the loss of fluidity of the melt
 - Facilitates top feeding of the melt
 - Avoids misrun
- P, R
 - Q, S
 - R, S
 - P, S

28. A 10 mm thick steel bar is to be horizontally cast with two correctly spaced top risers of adequate feeding capacity. Assuming end effect without chill, what should be the theoretical length of the bar ?
(GATE-PI-02)

- (a) 96 mm
(b) 132 mm
(c) 192 mm
(d) 156 mm

29. Match the process with mold making techniques.
(GATE-PI-02)

Process	Mold making technique			
P. Green sand molding	1. Pouring			
Q. Shell molding	2. Dipping			
R. Investment molding	3. Compaction			
S. Ceramic molding	4. Resin bonding			

	P	Q	R	S
(a)	2	1	3	4
(b)	1	3	4	2
(c)	4	2	1	3
(d)	3	4	2	1

30. Gating ratio of 1: 2: 4 is used to design the gating system for magnesium alloy casting. This gating ratio refers to the cross section areas of the various gating elements as given below (GATE-PI-03)

- Down sprue
- Runner
- Ingates

The sequence of the above elements in the ratio 1:2:4 is

- (a) 1, 2 & 3
(b) 1, 3 & 2
(c) 2, 3 & 1
(d) 3, 1 & 2

31. A casting of size 100 mm × 100 mm × 50 mm is required. Assume volume shrinkage of casting as 2.6%. If the height of the riser is 80 mm and riser volume desired is 4 times the shrinkage in casting, what is the appropriate riser diameter in mm?
(GATE-PI-03)

- (a) 14.38
(b) 20.34
(c) 28.76
(d) 57.52

32. Chaplets are placed between mould and core surfaces in order to:
(GATE-PI-03)

- Reduce directional solidification
- Help local alloying of molten metal
- Help easy removal of core from casting
- Prevent core movement due to buoyancy

33. Match the items of List-I (Equipment) with the items of List-II (Process) and select the correct answer using the given codes (GATE-PI-04)

List-I (Equipment)	List-II (Process)
P. Hot Chamber Machine	1. Cleaning
Q. Muller	2. Core making
R. Dielectric Baker	3. Die Casting
S. Sand Blaster	4. Annealing
	5. Sand mixing

	P	Q	R	S
(a)	2	1	4	5
(b)	4	2	3	5
(c)	4	5	1	2
(d)	3	5	2	1

34. The shape factor for a casting in the form of an annular cylinder of outside diameter 30 cm, inside diameter 20 cm and height 30 cm (correction factor $k = 1.0$) will be (GATE-PI-05)

- (a) 21.77 (b) 6.28 (c) 9.42 (d) 12.28

35. A cast steel slab of dimension 30×20×5 cm is poured horizontally using a side riser. The riser is cylindrical in shape with diameter and height, both equal to D. The freezing ratio of the mould is (GATE-PI-05)

- (a) 8D / 75 (b) 4D / 75
(c) 75 / 8D (d) 75 / 4D

36. A casting of size 400 mm × 200 mm × 140 mm solidifies in 20 min. The solidification time for a casting 400 mm × 200 mm × 35 mm under similar conditions is (GATE-PI-06)

- (a) 2 min (b) 3.3 min (c) 4 min (d) 8.1 min

37. Match the following. (GATE-PI-07)

Group-I
(Casting process)

P. Sand casting

Q. Centrifugal casting

R. Investment casting

S. Die casting

Group – II
(Application)

1. Turbine blades

2. I.C engines

3. Large bells

4. Pulleys

	P	Q	R	S
(a)	4	1	3	2
(b)	2	4	3	1
(c)	3	4	1	2
(d)	3	2	1	4

Statement for Linked Answer Q.Nos. 38 & 39

In a sand casting process, a sprue of 10 mm base diameter and 250 mm height leads to a runner which fills a cubical mold cavity of 100 mm size. (GATE-PI-07)

 38. The volume flow rate (in mm^3/s) is

- (a) 0.8×10^5 (b) 1.1×10^5
 (c) 1.7×10^5 (d) 2.3×10^5

39. The mold filling time in seconds is (GATE-PI-07)

- (a) 2.8 (b) 5.78 (c) 7.54 (d) 8.41

40. In sand casting of hollow part of lead, a cylindrical core of diameter 120 mm and height 180 mm is placed inside the mould cavity. The densities of core material and lead are 1600 kg/m^3 and 11300 kg/m^3 respectively. The net force (in N) that tends to lift the core during pouring of molten metal will be (GATE-PI-08)

- (a) 19.7 (b) 64.5 (c) 193.7 (d) 257.6

41. A solid cylinder of diameter D and height equal to D and a solid cube of side L are being sand cast by using the same material. Assuming there is no superheat in both the cases, the ratio of solidification times of the cylinder to the solidification time of the cube is (GATE-PI-09)

- (a) $(L/D)^2$ (b) $(2L/D)^2$
 (c) $(2D/L)^2$ (d) $(D/L)^2$

42. During the filling process of a given sand mould cavity by molten metal through a horizontal runner of circular cross section, the frictional head loss of the molten metal in the runner will increase with the (GATE-PI-10)

(a) Increase in runner diameter
 (b) Decrease internal surface roughness of the runner
 (c) Decrease in length of runner
 (d) Increase in average velocity of molten metal

43. In a sand casting process, a sphere and a cylinder of equal volumes are separately cast from the same molten metal under identical conditions. The height and diameter of the cylinder are equal. The ratio of the solidification time of the sphere to that of the cylinder is (GATE-PI-11)

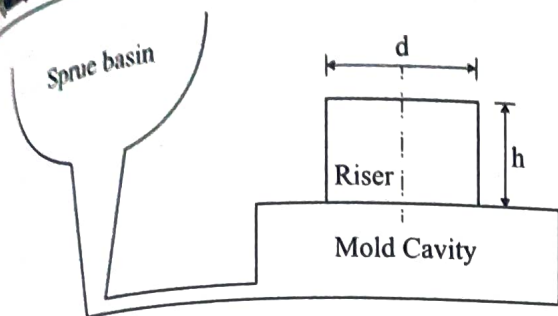
- (a) 1.14 (b) 0.87 (c) 1.31 (d) 0.76

44. A mold having dimensions $100 \times 90 \times 20$ (all in mm) is filled with molten metal through a gate with height 'h' and C.S area A, the mould filling time is t_1 . The height is now quadrupled and the cross sectional area is halved. The corresponding filling time is t_2 . The ratio t_2/t_1 is (GATE-PI-12)

- (a) $\frac{1}{\sqrt{2}}$ (b) 1 (c) $\sqrt{2}$ (d) 2

45. An aluminium alloy (density 2600 kg/m^3) casting is to be produced. A cylindrical hole of 100 mm diameter and 100 mm length is made in the casting using sand core (density 1600 kg/m^3). The net buoyancy force (in Newton) acting on the core is (GATE-ME-14-SET-1)

46. A cylindrical blind riser with diameter 'd' and height 'h', is placed on the top of the mold cavity of a closed type sand mold as shown in the figure. If the riser is of constant volume, then the rate of solidification in the riser is the least when the ratio h:d is (GATE-ME-14-SET-3)



- (a) 1:2 (b) 2:1 (c) 1:4 (d) 4:1

47. A cylindrical riser of 6 cm diameter and 6 cm height has to be designed for a sand casting mould for producing a steel rectangular plate casting of 7 cm × 10 cm × 2 cm dimensions having the total solidification time of 1.36 minute. The total solidification time (in minute) of the riser is _____
(GATE-ME- 14-SET-4)

48. For a given volume of a riser, if the solidification time of the molten metal in riser needs to be quadrupled, the surface area of the riser should be made _____
(GATE-PI- 14)

- (a) one-fourth (b) half
(c) double (d) four times

49. The solidification time of a casting is proportional to $\left(\frac{V}{A}\right)^2$, Where V is the volume of the casting and A is the total casting surface area losing heat. Two cubes of same material and size are cast using sand casting process. The top face of one of the cubes is completely insulated. The ratio of the solidification time for the cube with top face insulated to that of the other cube is _____
(GATE -15 -Set 1)

- (a) $\frac{25}{36}$ (b) $\frac{36}{25}$ (c) 1 (d) $\frac{6}{5}$

50. A cube and a sphere made of cast iron (each volume 1000 cm³) were cast under identical conditions. The time taken for solidifying the cube was 4 s. The solidification time (in s) for the sphere is _____
(GATE -15 -Set 2)

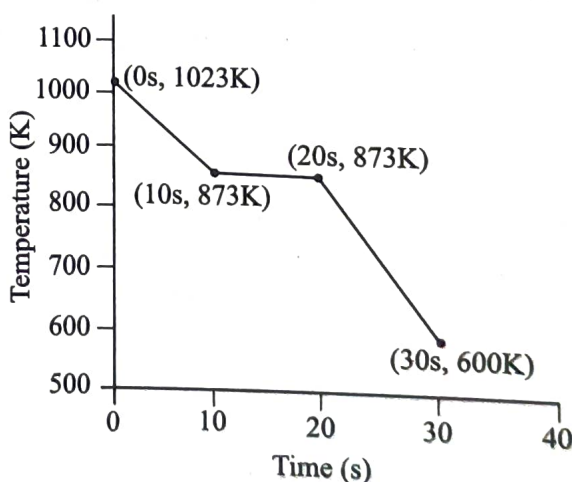
51. The dimensions of a cylindrical side riser (height = diameter) for a 25 cm × 15 cm × 5 cm steel casting are to be determined. For the tabulated shape factor values given below, the diameter of the riser (in cm) is _____
(GATE -15 -Set 3)

Shape factor	2	4	6	8	10	12
Riser volume/ Casting volume	1.0	0.70	0.55	0.50	0.40	0.35

52. Ratio of solidification time of a cylindrical casting (height = radius) to that of a cubic casting of side two times the height of cylindrical casting is _____
(GATE -15 -Set 3)

53. In a metal casting process, molten copper alloy is poured into a sand mould. The level of molten metal in the pouring basin is at a height of 300 mm from the runner having diameter of 10 mm. If the density and melting temperature of molten copper alloy are 9000 kg/m³ and 1000°C respectively, then the rate of flow of molten metal into the mould neglecting friction and other losses, in cm³/s, is _____
(GATE - PI -15)

54. Heat is removed from a molten metal of mass 2 kg at a constant rate of 10 kW till it is completely solidified. The cooling curve is shown in the figure.



Assuming uniform temperature throughout the volume of the metal during solidification, the latent heat of fusion of the metal (in kJ/kg) is _____
(GATE - 16 - SET - 1)

55. A cylindrical job with diameter of 200 mm and height of 100 mm is to be cast using modulus method of riser design. Assume that the bottom surface of cylindrical riser does not contribute as cooling surface. If the diameter of the riser is equal to its height, then the height of the riser (in mm) is

(GATE - 16 - SET - 1)

- (a) 150 (b) 200 (c) 100 (d) 125

56. Gray cast iron blocks of size 100 mm × 50 mm × 10 mm with a central spherical cavity of diameter 4 mm are sand cast. The shrinkage allowance for the pattern is 3%. The ratio of the volume of the pattern to volume of the casting is _____

(GATE - 16-SET-2)

57. Two cast iron blocks P and Q, each of 500 mm length, have the same cross-sectional area. Block P has rectangular cross-section of 100 mm × 200 mm. Block Q is of square cross-section. Both P and Q were cast under the same conditions with all their surfaces enclosed within the mould. The solidification time of a casting is proportional to the square of the ratio of its volume to its surface area. The ratio of solidification time of the block P to that of the block Q is _____. (GATE - PI-16)

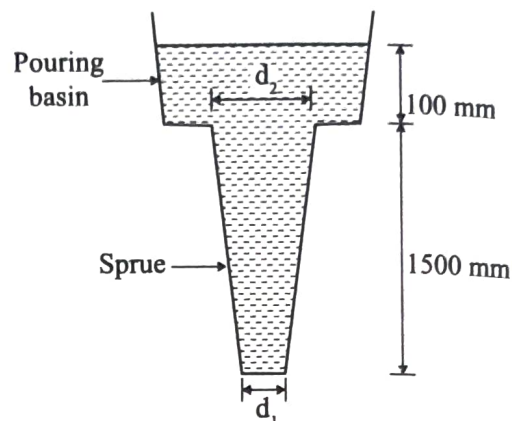
58. A 300 mm long copper wire of uniform cross-section is pulled in tension so that a maximum tensile stress of 270 MPa is developed within the wire. The entire deformation of the wire remains linearly elastic. The elastic modulus of copper is 100 GPa. The resultant elongation (in mm) is ____.

(GATE - PI-16)

59. A sprue in a sand mould has a top diameter of 20 mm and height of 200 mm. The velocity of the molten metal at the entry of the sprue is 0.5 m/s. Assume acceleration due to gravity as 9.8 m/s² and neglect all losses. If the mould is well ventilated, the velocity (upto 3 decimal points accuracy) of the molten metal at the bottom of the sprue is _____ m/s.

(GATE - 17 - SET - 1)

60. Schematic diagram of pouring basin and sprue of a gating system is shown in the Figure. Depth of molten metal in the pouring basin is 100 mm and the height of the sprue is 1,500 mm.



Considering the cross-section of the sprue is circular, the ratio $d_1:d_2$ to avoid aspiration is

(GATE - PI - 17)

- (a) 3:2 (b) 5:6 (c) 15:16 (d) 1:2

61. For sand-casting a steel rectangular plate with dimensions 80 mm × 120 mm × 20 mm, a cylindrical riser has to be designed. The height of the riser is equal to its diameter. The total solidification time for the casting is 2 minutes. In Chvorinov's law for the estimation of the total solidification time, exponent is to be taken as 2. For a solidification time of 3 minutes in the riser, the diameter (in mm) of the riser is _____ (correct to two decimal places).

(GATE - 18 - SET - 2)

62. A pressure die casting set-up was tested by injecting water (density 1000 kg/m³) at a pressure of 200 bar. Mould-filling time was found to be 0.05 s. Afterwards, the actual casting is made by injecting the liquid metal (density 2000 kg/m³) at an injection pressure of 400 bar. Neglect all losses (friction, viscous-effect, etc). The approximate mould-filling time (in s) is

(PI_GATE - 18)

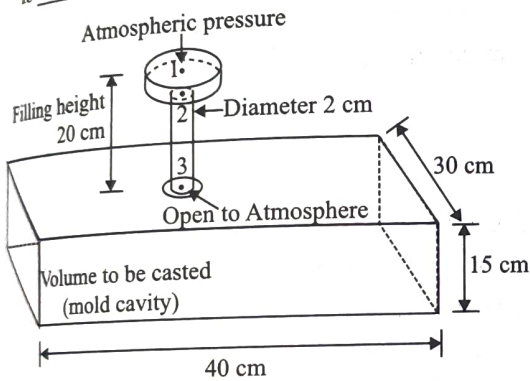
- (a) 0.05 (b) 0.075 (c) 0.1 (d) 0.2

63. Match the following sand mold casting defects with their respective causes. (GATE - 19 - SET - 1)

	Defect		Cause
P.	Blow hole	1.	Poor collapsibility
Q.	Misrun	2.	Mold erosion
R.	Hot tearing	3.	Poor permeability
S.	Wash	4.	Insufficient fluidity

- (a) P-3, Q-4, R-1, S-2 (b) P-4, Q-3, R-1, S-2
(c) P-2, Q-4, R-1, S-3 (d) P-3, Q-4, R-2, S-1

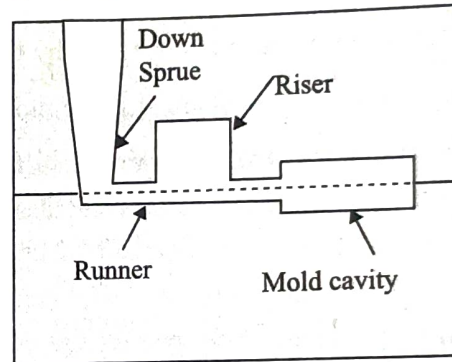
64. The figure shows a pouring arrangement for casting of a metal block. Frictional losses are negligible. The acceleration due to gravity is 9.81 m/s^2 . The time (in s, round off to two decimal places) to fill up the mold cavity (of size $40 \text{ cm} \times 30 \text{ cm} \times 15 \text{ cm}$) is _____. (GATE - 19 - SET - 2)



65. A sand casting process has a mold constant of 2 s/mm^2 and solidification exponent of 2. If the solidification time is to be doubled for a given unit volume of material, the corresponding reduction in the cast surface area (in %) is _____. (GATE-19_PI)
66. True centrifugal casting process in horizontal configuration is to be using for casting a metallic cylinder with outside diameter 0.275 m and inside diameter 0.250 m. If G-factor (ratio of centrifugal force experienced by the rotating cast metal to its weight) is 65 and acceleration due to gravity is 9.8 m/s^2 , the minimum rotational speed (in rpm) required is closest to _____. (GATE - 19_PI)
- (a) 325 (b) 650 (c) 975 (d) 1300

67. A mould cavity of 1200 cm^3 volume has to be filled through a sprue of 10 cm length feeding a horizontal runner. Cross-sectional area at the base of the sprue is 2 cm^2 . Consider acceleration due to gravity as 9.81 m/s^2 . Neglecting frictional losses due to molten metal flow, the time taken to fill the mould cavity is _____ seconds. (round off to 2 decimal places). (GATE-20-SET-2)

68. To manufacture a product by casting, molten metal is poured in a cavity of rectangular cross-section in a sand mold with a side blind riser as shown in the figure. The dimensions of the mold cavity are $60 \text{ cm} \times 40 \text{ cm} \times 20 \text{ cm}$.



The riser is cylindrical in shape with diameter equal to height. It is required that the solidification time of the riser should be 25% greater than that of the mold. Using Chvorinov's rule, the diameter of the riser (rounded off to one decimal place) in cm should be _____. (GATE-PI-20)

69. A true centrifugal casting operation needs to be performed horizontally to make copper tube sections with outer diameter of 250 mm and inner diameter of 230 mm. The value of acceleration due to gravity, $g = 10 \text{ m/s}^2$. If a G-factor (ratio of centrifugal force to weight) of 60 is used for casting the tube, the rotational speed required is _____ rpm (round off to the nearest integer). (GATE-21_SET-1)

70. A cast product of a particular material has dimensions 75 mm x 125 mm x 20 mm. The total solidification time for the cast product is found to be 2.0 minutes as calculated using Chvorinov's rule having the index, $n = 2$. If under the identical casting conditions, the cast product shape is changed to a cylinder having diameter = 50 mm and height = 50 mm, the total solidification time will be _____ minutes (round off to two decimal places).

(GATE-21_SET-2)

71. A company is producing a disc-shaped product of 50 mm thickness and 1.0 m diameter using sand casting process. The solidification time of the above casting process is estimated by Chvorinov's equation $= B \left[\frac{V}{A} \right]^2$, where B is the mold constant, and V and A are the volume and surface area of the casting, respectively. It is decided to modify both the thickness and diameter of the disc to 25 mm and 0.5 m, respectively, maintaining the same casting condition. The percentage reduction in solidification time of the modified disc as compared to that of the bigger disc is _____.

[round off to one decimal place] (GATE_PI-21)

72. The solidification of a cubical casting of side 100 mm takes place with volumetric solidification shrinkage and solid contraction of 10 % each. The shape of the casting is retained on cooling to room temperature. The side of the cubical cast, in mm, at room temperature is _____ (round off to 2 decimal places).

(GATE_PI-23)

73. A cylindrical casting has 10 cm diameter and a mass of 12.56 kg. The material density is $7.85 \times 10^{-3} \text{ kg/cm}^3$. The value of exponent 'n' is 2 and solidification time is 12 min. The Chvorinov's constant, in min/cm^2 , is _____ (round off to 2 decimal places).

(GATE_PI-23)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (a&c)	02. (c)	03. (a)	04. (a)	05. (c)
06. (a)	07. (d)	08. (c)	09. (b,c)	10. (c)
11. (b)	12. (d)	13. (c)	14. (d)	15. (c)
16. (c)	17. (b)	18. (d)	19. (d)	20. (a)
21. (d)	22. (d)	23. (c)	24. (*)	25. (b)
26. (b)	27. (c)	28. (*)	29. (a)	30. (c)
31. (a)	32. (*)	33. (*)	34. (c)	35. (*)
36. (a)	37. (d)	38. (*)	39. True	40. False
41. (d)	42. (a)	43. (d)	44. (d)	45. (a)
46. (d)	47. (c)	48. (a)	49. (b)	50. (b)
51. (b)	52. (c)	53. (a)	54. (a)	55. (b)
56. (c)	57. (b)	58. (c)	59. (c)	60. (b)
61. (b)	62. (b)	63. (c)	64. (a,b)	65. (a)
66. (a,b,d)				

TWO MARK QUESTIONS

01. (c)	02. (*)	03. (*)	04. (*)	05. (b)
06. (d)	07. (d)	08. (b)	09. (b)	10. (b)
11. (d)	12. (c)	13. (b)	14. (d)	15. (a)
16. (a)	17. (a)	18. (c)	19. (*)	20. (a)
21. (*)	22. (*)	23. (c)	24. (*)	25. (a)
26. (d)	27. (b)	28. (b)	29. (d)	30. (a)
31. (d)	32. (d)	33. (d)	34. (a)	35. (a)
36. (b)	37. (c)	38. (c)	39. (b)	40. (c)
41. (d)	42. (d)	43. (c)	44. (b)	45. 7.7
46. (a)	47. 3	48. (b)	49. (b)	50. 6.16
51. 10.5 to 10.7		52. 0.5625	53. 185 to 195	
54. 50	55. (a)	56. 1.0927	57. 0.9016	
58. 0.81 mm		59. 2.042	60. (d)	61. 51.84
62. (a)	63. (a)	64. 28.92	65. 29.3	66. (b)
67. 4.28	68. 36.6	69. 662	70. 2.82	71. 75
72. 93.21	73. 3			

One Mark Solutions

01. Ans: (a & c)

Sol: The riser should be designed in such way that it should supply molten metal to compensate liquid shrinkage. This can be achieved only when the molten metal in the riser solidifies after or at the same time of the casting.

02. Ans: (c)

Sol: The shrinkage allowance provided on the pattern is to compensate the solid shrinkage taking place during cooling of material from freezing temperature to room temperature as a solid means that it is for compensating solid shrinkage.

03. Ans: (a)

Sol: By providing chills (high thermal conductivity metal pieces) it is possible to direct heat to transfer in a required direction called as directional solidification.

04. Ans: (a)

Sol: By adding more water into the molding sand, the flowability of moulding sand increases so that it is possible to make complex cavity shapes.

05. Ans: (c)

Sol: Maximum pressure occurs in the gating system where the ratio of cross sectional area of in gate to sprue is maximum.

Option	Ratio(A_g/A_s)	Remark
A	3/4	-
B	3/1	-
C	4/1	Maximum
D	1	-

Among above all options option (c) has maximum ratio.

06. Ans: (a)

$$\text{Sol: Shrinkage allowance on the pattern} = L \alpha (\Delta T) \\ = L \alpha (T_f - T_r)$$

Where,

T_f = freezing temperature

T_r = room temperature

L = Dimensions of component

α = Coefficient of thermal expansion

From the above equation, shrinkage allowance depends on dimension of the part, coefficient of thermal expansion, freezing temperature and then room temperature. Hence the only parameter which is not influencing is pouring temperature.

07. Ans: (d)

Sol: By using converging passage for the sprue with straight tapered or parabolic tapered will eliminate the aspiration effect.

08. Ans: (c)

Sol: With uniform ramming action, the mould will not get distorted and hence the casting is dimensionally stable.

09. Ans: (b & c)

Sol: Both are the correct answers, due to centrifugal force the compaction of the molten metal particles will take place which avoids the porosity, density becomes high and due to fast cooling fine grain structure will take place.

10. Ans: (c)

Sol: The bottom part of the skim bob is sometimes called as bottom well as it prevents only heavy impurities. Strainer may remove lighter impurities to some extent but skim bob separates all the lighter impurities.

11. Ans: (b)

Sol: Out of all the materials brass has the highest coefficient of thermal expansion hence it requires highest shrinkage allowance to be provided on the pattern.

12. Ans: (d)

Sol: In case of polystyrene patterns, the pattern will not be removed from the mold, but the hot molten metal is directly poured so that the molten metal will melt and evaporate the polystyrene pattern. In this way polystyrene will be removed from cavity through porosity property of mold or using through riser.

13. Ans: (c)

Sol: In casting process the solid shrinkage is compensated by providing shrinkage allowance on the pattern. The solid shrinkage is the shrinkage taking place during cooling of material from freezing temperature to the room temperature in solid state.

14. Ans: (d)

Sol: In centrifugal casting the centrifugal force acting on the particles depends on the density of the particles. The lighter impurities have lower density and so they are collected at the center.

15. Ans: (c)

Sol: Sprue is the connecting passage between the pouring basin and runner for supplying molten metal to the casting cavity. It is always vertical with straight tapered circular cross section.

16. Ans: (c)

Sol: To increase the hardness of green sand mold it is necessary to apply large amount of ramming force so that the porosity property of the mold will come down which reduces the permeability of the mold.

17. Ans: (b)

Sol: Misrun is the defect produced due to non-availability of molten metal for filling of farthest point from pouring point. This happens when the solidification of molten metal is started before complete filling of the molten metal into casting cavity. This can be eliminated by either increasing the pouring temperature of molten metal or by reducing pouring time. The pouring time is reduced by increasing the fluidity of molten metal.

18. Ans: (d)

Sol: The expendable pattern like wax pattern is used in the investment casting.

19. Ans: (d)

Sol: In hot chamber die casting only low melting point materials like lead, tin, zinc, cadmium upto aluminum can be casted. Out of the given materials only tin has less melting point than aluminum.

20. Ans: (a)

Sol: Cold shut is an interface in a casting that lacks complete fusion, because of the meeting of two streams of liquid metal from different gates.

21. Ans: (d)

Sol: The molding sand which contains moisture is called green sand mold.

22. Ans: (d)

Sol: Shake allowance is the only allowance which decreases the dimensions of the pattern. Hence called as negative allowance.

23. Ans: (c)

Sol: Slush casting is the open casting process in which the liquid molten metal is poured into the cavity, allow it to solidify for some time, invert the mold so that the un-solidified molten metal will fall down

and solidified metal retains in the mold. Now by breaking the mold the casting can be taken out. This is generally used for producing decorative parts like lamp shades.

24. **Sol:** Distortion or bending type of defect is produced in the casting.

25. **Ans: (b)**
Sol: Investment casting is used for producing complex shapes of the components like turbine blades.

26. **Ans: (b)**
Sol: By varying the force applied during compaction stage of powder metallurgy process, the density of the product can be varied.

27. **Ans: (c)**
Sol: The strength of the part obtained after the compacting stage is called green strength.

28. **Ans:** Metals, without chill block.

29. **Ans: (a)**
Sol: Because of usage of converging passage only the aspiration effect will be eliminated so that the defect produced due to aspiration effect like blow holes can be eliminated.

30. **Ans: (c)**
Sol: More fluidity of casting metal means that viscosity is less and hence flow rate increases which reduces the pouring time.

31. **Ans: (a)**
Sol: In investment casting to make the mould, the prepared pattern is dipped into a slurry made by suspending fine ceramic materials in a liquid. Such

as ethyl silicate or sodium silicate.

32. **Blow holes**

Sol: Because of low permeability of molds the air cannot escape from the mold and hence blow holes will be formed.

33. **Ans:** A. Lower, B. Higher

34. **Ans: (c)**

Sol: With increase in carbon content in plain carbon steels, the tensile strength and hardness of the material will increase.

35. **Ans:** A. Wax pattern, B. intricate shapes

36. **Ans: (a)**

Sol: Riser is acting as a reservoir for supplying molten metal to casting cavity for compensating liquid shrinkages during solidification.

37. **Ans: (d)**

Sol: Chaplets are used as additional supports for supporting the core to prevent movement of core due to buoyancy.

38.

Ans: Wax, Intricate

39. **Ans: True**

Sol: Clay binders are most commonly used binding agents mixed with the molding sands to provide the strength. The most popular clay types are kaolinite and bentonite. Amongst the two, the bentonite can absorb more water which increases its bonding power.

40. **Ans: False**

Sol:

- In shell moulding process, the sand mixed with a thermosetting resin is allowed to come into contact with a heated metallic pattern.

- In investment casting, the mould is prepared around an expendable pattern. The molten wax is used as a pattern material and is injected under pressure into a metallic die. To make the mould, the prepared pattern is dipped into a slurry made by suspending fine ceramic materials in a liquid such as ethyl silicate or sodium silicate.

41. Ans: (d)

Sol: Fluidity is the flow property of liquids but for defining the flow property of solids we use the word called as "flow ability", hence solid fluidity is not the property of molding sand.

42. Ans: (a)

Sol: Generally wood flour is added to molding sand as an additive to improve collapsibility and porosity property of molding sand.

43. Ans: (d)

Sol: The expendable pattern like wax pattern is used in the investment casting

44. Ans: (d)

Sol: Refer to the solution of Q.No. 20.

45. Ans: (a)

Sol: Due to centrifugal force, density of part is highest at the outer surface.

46. Ans: (d)

Sol: Hot chamber die casting is used for producing the castings which have very low melting point like lead, tin and zinc.

47. Ans: (c)

Sol: As the thermal diffusivity increases the thermal conductivity increases, so that rate of heat transfer increases and hence solidification time reduces.

48. Ans: (a)

Sol: In liquids, with increase of temperature the viscosity reduces which means fluidity increases.

49. Ans: (b)

Sol: The expendable pattern like wax pattern is used in the investment casting.

Investment casting requires a pattern for each mold produced and is an example of an expendable-mold, expendable-pattern process.

50. Ans: (b)

Sol: Let a_1 = side of cube

$$V_1 = \text{Volume of 1st casting} = a_1^3$$

$$\tau_1 = \text{solidification time for the 1st casting} = 5 \text{ min}$$

$$\text{Given } V_2 = 8V_1 = a_2^3 \dots \dots \dots \text{say}$$

$$a_2^3 = 8a_1^3 \Rightarrow a_2 = 2a_1$$

$$\frac{\tau_2}{\tau_1} = \left(\frac{M_2}{M_1} \right)^2 = \left(\frac{\frac{a_2}{6}}{\frac{a_1}{6}} \right)^2 = \left(\frac{a_2}{a_1} \right)^2 = 2^2 = 4$$

$$\tau_2 = 4, \Rightarrow \tau_1 = 4 \times 5 = 20 \text{ min}$$

51. Ans: (b)

Sol:

- Hot tears – differential cooling (Different parts of a component is cooled at different rates).
- Shrinkage– reduction in volume or dimensions.
- Blow hole – due to low porosity property of molding sand.
- Cold shut – due to improper mixing of two streams of molten metal.

52. Ans: (c)

Sol: Hot tears – differential cooling rate (different parts of a component are cooled at different rates). Also if the range of solidification temperature is high or wide, solidified metal will allow the conduction heat transfer and liquid will allow the convection, hence the cooling rates will become different.

53. Ans: (a)
Sol: The basic purpose of using the chills is to produce directional solidification.

54. Ans: (a)
Sol: Full mold casting is an evaporative pattern casting process. This process is a combination of sand casting and lost foam casting. It uses an *expanded polystyrene foam pattern* which is then surrounded by sand, much like sand casting. The metal is then poured directly into the mould, which vaporizes the foam upon contact.

55. Ans: (b)
Sol: Sprue feeds molten metal from pouring basin to gate.
Riser supplies molten metal to compensate for liquid shrinkage.
Gate regulates flow of molten metal into mould cavity.
Pouring basin acts as a reservoir for molten metal.

56. Ans: (c)
Sol: Rate of pouring of molten metal depends on the flow rate of molten metal. This depends on the choke area and it is the minimum area out of the cross sectional areas of sprue, runner and ingate.

57. Ans: (b)
Sol: In submerged arc welding, the arc is completely submerged inside the granular flux powder and forming as blanket.

58. Ans: (c)
Sol: Solidification time is inversely proportional to diffusivity and based on the given values diffusivity is highest for copper, next is aluminium and then steel. Hence the solidification time is lowest for copper, next aluminium and then steel.

59. Ans: (c)
Sol: When carbon dioxide is supplied to the dry sand it is chemically reacting with *sodium silicate* and produces silica gel which is a paste like material and on drying it gives high strength to the mold.

60. Ans: (b)
Sol: In pure metals, the metal cools rapidly at the mould walls, rapid cooling produces a solidified skin of fine equiaxed grains which are randomly-oriented small crystals.

61. Ans: (b)
Sol: A sprue, through which the molten metal flows downward.

62. Ans: (b)
Sol: Fluidity increases with increase in degree of super heat.

63. Ans: (c)
Sol:

- In permanent-mold casting (also called hard-mold casting), two halves of a mold are made from materials with high resistance to erosion and thermal fatigue, such as cast iron, steel, bronze, graphite, or refractory metal alloys. Typical parts made are automobile pistons, cylinder heads, connecting rods, gear blanks for appliances, and kitchenware. Deep drawing is used make low carbon oil pans.
- Powder metallurgy is used for manufacturing products or articles from powdered metals by placing these powders in molds and are compacting the same using heavy compressive force. Typical examples of such article or products are grinding wheels, filament wire, magnets, welding rods, tungsten carbide cutting tools, self-lubricating bearings electrical contacts and turbines blades having high temperature strength.

- Blow molding (or moulding) is a manufacturing process for forming and joining together hollow plastic parts. It is also used for forming glass bottles or other hollow shapes.

64. Ans: (a, b)

Sol: Riser should take more time for solidification than casting $\left(\frac{V}{A}\right)_r \geq \left(\frac{V}{A}\right)_c$

For top riser bottom area is not cooling surface and it is given that top cross section is also insulated. So only lateral area is cooling area.

$$\frac{\frac{\pi}{4} D^2 H}{\pi D H} \geq \frac{a^3}{6a^2} \Rightarrow \frac{D}{4} \geq \frac{a}{6} \Rightarrow D \geq \frac{2a}{3}$$

$\therefore$ Radius $R \geq \frac{a}{3}$ only answer is available $\frac{a}{2}$

(OR)

From Modulus Method :

$$M_r = 1.2 M_c$$

$$\frac{\frac{\pi}{4} D^2 H}{\pi D H} = 1.2 \frac{a^3}{6a^2} \Rightarrow \frac{D}{4} = 1.2 \frac{a}{6}$$

$R = 1.2 \times \frac{a}{3}$ This is the maximum radius required so only one multiple choice above it is $\frac{a}{2}$

65. Ans: (a)

Sol:

- Continuous casting, or strand casting, was first developed for casting non-ferrous metal strip and is now widely used for steel production and at low cost.
- * In strip casting thin *slabs or strips* are produced from molten metal, which solidifies in manner similar to strand casting, but the hot solid is then rolled to form the final shape.
- Investment casting enables precise fabrication of relatively small components with intricate shape and excellent surface finish and dimensional accuracy.

- In centrifugal casting process, the mold is rotated rapidly about its central axis while the metal is poured into it. Owing to the centrifugal force, a continuous pressure will act on the metal as it solidifies. The slag, oxides and other inclusions being lighter get separated from the metal and segregate towards the center. This process is normally used for the *making hollow pipes, tubes, hollow bushes, etc., which are axisymmetric with a concentric hole*. Since the metal is always pushed outward because of the centrifugal force, no core needs to be used for making the concentric hole.

66. Ans: (a, b and d)

Sol: In a metal casting process to manufacture parts, both patterns and moulds provide shape by dictating where the material should or should not go.

- Mould walls indicate boundaries within which the molten part material is allowed, while pattern walls indicate boundaries of regions where mould material is not allowed.
- Moulds can be used to make patterns.
- Patterns can be used to make moulds.

Two Marks Solutions

01. Ans: (c)

Sol: Solidification time $(\tau) \propto M^2$

where, M (modulus) = $\frac{\text{volume}}{\text{surface area}}$

$$\text{For cube } M = \frac{a^3}{6a^2} = \frac{a}{6}$$

a = side of the cube

$$\frac{\tau_1}{\tau_2} = \left(\frac{M_1}{M_2}\right)^2 = \left(\frac{a_1}{a_2}\right)^2 = \left(\frac{2}{4}\right)^2 = \frac{1}{4}$$

$$\Rightarrow \tau_2 = \tau_1 \times 4 = 2 \times 4 = 8 \text{ min}$$

02. Ans: A-4, B-3, C-6, D-5

Sol:

- Slush casting is mainly used for producing decorative parts like lamp shades, X-mas trees etc.
- Shell molding is used for producing circular and symmetrical jobs only (cylinder block).
- Dry sand molding is for CI brake shoe and centrifugal casting is used for producing large size hollow cylindrical parts without using core.

03. Ans: A-2, B-6, C-5, D-3

Sol: In the given question only 4 of the casting processes will match with the products. Always connecting rods are made by forging operation, turbine blades are made by investment casting, cast iron pipes by centrifugal casting and carburetor body is made by pressure die casting.

04. Ans: A-5, B-3, C-1, D-2

Sol:

- In sand casting because of chances of compression of molding sand parts may warp.
- In plaster mold casting due to good surface finish no post casting process like machining is required.
- Shell molding is used for circular and symmetrical shapes only and due to high thermal conductivity of the mold cooling rate is high and hence surface hardening treatment will take place which gives

hard surface and soft interior.

05. Ans: (b)

Sol: Solidification time $(\tau) = KM^2$

where, K = solidification factor, M = modulus

$$\text{For sphere } (M) = \frac{\text{Volume}}{\text{surface area}} = \frac{\frac{4}{3}\pi R^3}{4\pi R^2}$$

$$M = \frac{R}{3} = \frac{D}{6}$$

$$\tau = 0.97 \times 10^6 \times \left(\frac{D}{6}\right)^2$$

$$= 0.97 \times 10^6 \times \left(\frac{0.2}{6}\right)^2 = 1078 \text{ sec}$$

06. Ans: (d)

Sol: $\frac{\text{Volume of pattern}}{\text{Volume of casting}}$

$$= \frac{(1.01 \times 200)(1.01 \times 100)(1.01 \times 10)}{200 \times 100 \times 10} = 1.03$$

Note: Solid shrinkage is dimensional shrinkage where as liquid shrinkage is volumetric shrinkage.

07. Ans : (d)

Sol:

- In casting always Gravity die casting is used for producing simple parts like IC engine piston made by aluminum alloy.
- Pressure die casting is used for producing complex shapes like carburetor body made by aluminum alloy.

08. Ans: (b)

Sol: Length of sprue = 20 cm

$$\text{C.S area of Base of sprue} = 1 \text{ cm}^2 = A_3 = A_c$$

Volume of casting = 1000 cc

$$\text{Time required for filling} = \frac{\text{Volume}}{A_c \times V}$$

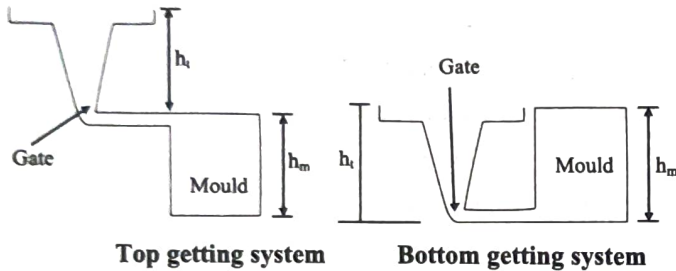
$$= \frac{1000}{1 \times \sqrt{2gh}} = \frac{1000}{1 \times \sqrt{2 \times 981 \times 20}} = 5.05 \text{ sec}$$

09. Ans: (b)

Sol: Time taken to fill the mould with top gate,

$$t_A = \frac{A_m h_m}{A_g \sqrt{2gh_t}} \quad \dots\dots (1)$$

Where, A_m = area of mould, h_m = height of mould
 A_g = area of gate, h_t = total height



Given that, $h_t = h_m$, so, eq.(1) becomes

$$t_A = \frac{A_m \sqrt{h_t}}{A_g \sqrt{2g}} \quad \dots\dots (2)$$

Time taken to fill the mould with bottom gate,

$$t_B = \frac{2A_m}{A_g \sqrt{2g}} \times (\sqrt{h_t} - \sqrt{h_t - h_m})$$

$$t_B = \frac{2A_m}{A_g \sqrt{2g}} \times \sqrt{h_t} \quad (\because h_t = h_m) \quad \dots\dots (3)$$

From eq. (ii) and (iii), we get

$$\frac{t_B}{t_A} = \frac{2}{1}$$

$$\Rightarrow t_B = 2 t_A$$

10. Ans: (b)

Sol: Steels with carbon percentage less than 0.8% are called hypo-eutectoid steels and above 0.8% is called hyper-eutectoid steels.

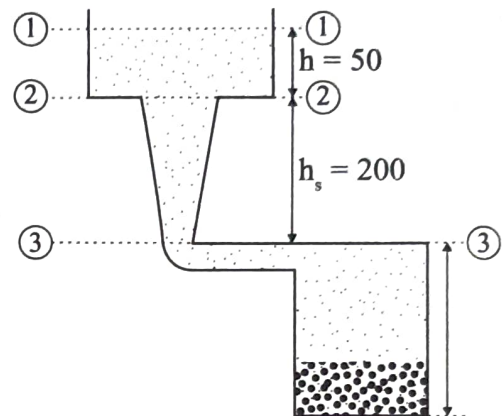
11. Ans: (d)

$$\text{Sol: } \frac{\tau_{\text{cube}}}{\tau_{\text{sp}}} = \left(\frac{(V/A_s)_{\text{cube}}}{(V/A_s)_{\text{sp}}} \right)^2 = \left(\frac{(A_s)_{\text{sp}}}{(A_s)_{\text{cube}}} \right)^2$$

$$= \left(\frac{4\pi r^2}{6\ell^2} \right)^2 = \left(\frac{4\pi}{6} \right)^2 \left(\frac{r}{\ell} \right)^4$$

12. Ans: (c)

Sol: h = height of sprue = 200 mm; $A_2 = 650 \text{ mm}^2$
 Q = flow rate = $6.5 \times 10^5 \text{ mm}^3/\text{s}$; $g = 10^4 \text{ mm/sec}^2$



$$Q = A_2 V_2$$

$$V_2 = \frac{6.5 \times 10^5}{650} = 1000 \text{ mm/sec}$$

$$V_2 = \sqrt{2gh} \Rightarrow \sqrt{2 \times 10^4 \times h} = 1000 \text{ mm/sec}$$

$$h = 50 \text{ mm}$$

= height of molten metal in the pouring basin

h_t = total height of molten metal above the bottom of the sprue

$$= 200 + 50 \text{ mm}$$

$$Q = A_2 V_2 = A_3 V_3$$

$$6.5 \times 10^5 = A_3 \sqrt{2 \times 10^4 \times 250}$$

$$\Rightarrow A_3 = 290.7 \text{ mm}^2$$

13. Ans: (b)

Sol: Riser is compensating the liquid shrinkages taking place during solidification like liquid phase and liquid to solid phase.

$$\text{i.e., } 3\% + 4\% = 7\%$$

14. Ans: (d)

Sol:

- Chills are used for getting directional solidification.
- Chaplets are used for supporting the core to avoid the movement of core due to buoyancy.
- Riser is acting as a reservoir to supply molten metal to cavity for compensating liquid shrinkages taking place during solidification.



Exothermic pads are used for avoiding sand erosion and for controlling cooling of critical sections.

15. Ans: (a)

Sol: The given ratio is the gating ratio i.e., the ratio of C.S areas of bottom of sprue, runner and ingate.

16. Ans: (a)

Sol: $\frac{h}{r} = 2 \Rightarrow h = 2r$

$\therefore h = D$

Consider it to be taken as side riser

From the given condition

$$\tau_{sp} = \tau_{cyl} \Rightarrow \left(\frac{V}{A_s}\right)_{sp}^2 = \left(\frac{V}{A_s}\right)_{cyl}^2$$

$$\Rightarrow \left(\frac{D}{6}\right)_{sp}^2 = \left(\frac{D}{6}\right)_{cyl}^2$$

$$\Rightarrow D_{sp} = D_{cyl} = 100 \text{ mm} = 10 \text{ cm}$$

17. Ans: (a)

Sol: Side of cubical casting = 50 mm

V_c = Volume of cavity

$$= 50 \times 50 \times 50 = 125000 \text{ mm}^3$$

Volume of casting after solidification shrinkage

$$= 0.96 \times 125000$$

$$= 120000 \text{ mm}^3$$

Volume of casting after solid shrinkage

$$= 0.94 \times 120000$$

$$= 112800 \text{ mm}^3 = a^3$$

$$\Rightarrow a = (112800)^{0.333} = 48.32 \text{ mm}$$

Side of the cube after solidification and shrinkage

or contraction = 48.32 mm

18. Ans: (c)

Sol: Refer to the solution of Q.NO 13 of 1 Mark.

19. Ans: 217 cm/sec

Sol: $h_{act} = 30 \text{ cm}$

$$h_{with loss} = 0.8h_{act} = 0.8 \times 30 \text{ cm}$$

$$V = \sqrt{2gh_{with loss}} = \sqrt{2 \times 981 \times 30 \times 0.8} = 217 \text{ cm/sec}$$

20. Ans: (a)

Sol: Shrinkage allowance on the pattern

$$= L.\alpha.\Delta T$$

$$= L.\alpha.(T_f - T_r),$$

where T_f = freezing temperature

T_r = room temperature

From the above equation, shrinkage allowance depends on dimension of the part, coefficient of thermal expansion, freezing temperature and then room temperature. Hence the only parameter which is not influencing is pouring temperature.

21. Ans: A-4, B-5, C-1, D-2

Sol: *Sweep pattern*: It is used to sweep the complete casting by means of a plane sweep. These are used for generating large shapes, which are symmetrical or prismatic in nature. Such as *bell shaped or cylindrical*.

Cope and drag pattern: In machine moulding using squeeze-jolt machines cope can not be prepared by placing over the drag. So cope and drag pattern is used.

Follow board pattern is adopted for those castings where there are some portions, which are structurally weak and also having irregular parting lines.

Loose piece pattern is used when the contours of the part is such that, withdrawing the pattern from the mould is not possible.

22. Ans: A-3, B-5, C-4, D-2

Sol:

- Hot tears defect is caused due to high pouring temperature.
- Porosity defect is caused due to gas entrapment.
- Sand inclusions defect is caused due to loosely rammed sand mold.
- Shrinkage cavity defect is caused due to inadequate risering.

23. Ans: (c)

Sol: Solidification time (τ) $\propto (V/A_s)^2$.

24. Ans: A-3, B-5, C-4, D-2

Sol: Refer to the solution of Q. No. 22.

25. Ans: (a)

$$\begin{aligned} \text{Sol: Pouring time} &= \frac{\text{Volume}}{A_c \times V_{\max}} \\ &= \frac{10^6}{200 \times \sqrt{2 \times 10 \times 10^3 \times 175}} = 2.67 \text{ sec} \end{aligned}$$

26. Ans: (d)

$$\begin{aligned} \text{Sol: Permeability number} &= \frac{501.28}{PT} \\ &= \frac{501.28}{10 \times 1} = 50.128 \end{aligned}$$

27. Ans: (b)

Sol: Gating system will be designed such that

- the time taken for pouring the molten metal into the cavity must be as minimum as possible.
- it should avoid aspiration effect.
- it should not allow the impurities present in the molten metal.
- compensate the loss of fluidity of the melt.
- avoids misrun defects etc.

28. Ans: (b)

Sol: Length of casting plate with two risers with end wall effect and without chills is

$$L = 13 \times t = 13 \times 10 = 130 \text{ mm}$$

29. Ans: (d)

Sol:

- The required strength of green sand mould is obtained by compaction.
- In shell moulding the phenolic resin is used as mold material.
- In investment casting the pattern is dipped into a slurry of refractory materials such as a very fine silica and binders, including water, ethyle silicate and acids.

- In Ceramic Mold Casting: The slurry is a mixture of fine grained zircon, aluminium oxide and fused silica, which are mixed with bonding agents and poured over the pattern, which has been placed in flask.

30. Ans: (a)

Sol: Gating ratio is the ratio of cross sectional area of bottom of sprue, runner and ingate.

31. Ans: (d)

$$\text{Sol: } V_c = 100 \times 100 \times 50 = 5 \times 10^5 \text{ mm}^3$$

$$V_{sc} = \frac{2.6}{100} \times V_c = 13000 \text{ mm}^3$$

Riser height (H) = 80 mm

$$V_r = 4 V_{sc}$$

$$\frac{\pi}{4} D^2 H = 4 \times 13000$$

$$\Rightarrow D = \sqrt{\frac{4 \times 13000 \times 4}{\pi \times 80}} = 28.77 \text{ mm}$$

According to sufficient condition

$$\tau_r \geq \tau_c$$

$$M_r \geq M_c$$

$$\left(\frac{V}{A_s} \right)_r \geq \left(\frac{V}{A_s} \right)_c \Rightarrow D_r \geq D_c$$

$\therefore$ Based on this $D_r > 28.77$

$$\text{Hence } D_r = 57.52 \text{ mm}$$

32. Ans: (d)

Sol: Chaplets are used as additional supports for supporting the core to prevent movement of core due to buoyancy.

33. Ans: (d)

Sol: Hot chamber machine is used for die casting, muller is used for mixing sand, die electric baker is used for making the cores with bentonite as bonding material and sand blaster is used for cleaning.

36. Ans: (a)

Sol: In Castings shape factor (S.F) = $\frac{L+W}{t}$

L = height = 30 cm

$$W = \pi \times D_{\text{mean}} = \pi \times \frac{30+20}{2} = \pi \times 25 = 78.5 \text{ cm}$$

$$t = \frac{30-20}{2} = 5 \text{ cm}$$

$$\text{S.F} = \frac{30+78.5}{5} = 21.7$$

37. Ans: (a)

Sol: Casting size = $30 \times 20 \times 5 \text{ cm}$

Side riser, $D = H$

$$\left(\frac{V}{A_s}\right)_C = \frac{30 \times 20 \times 5}{2(30 \times 20 + 20 \times 5 + 5 \times 30)}$$

$$= \frac{3000}{1700} = \frac{30}{17}$$

$$\left(\frac{V}{A_s}\right)_R = \frac{D}{6}$$

$$\text{Freezing ratio} = \frac{(V/A_s)_R}{(V/A_s)_C} = \left(\frac{D/6}{30/17}\right)$$

$$= \left(\frac{D \times 17}{30 \times 6}\right) = 0.095 D \approx \frac{8D}{75}$$

38. Ans: (b)

$$\text{Sol: } V_1 = 400 \times 200 \times 140 = 112 \times 10^5 \text{ mm}^3$$

$$V_2 = 400 \times 200 \times 35 = 28 \times 10^5 \text{ mm}^3$$

$$A_{s1} = 2 \times (400 \times 200 + 200 \times 140 + 140 \times 400)$$

$$= 328000 \text{ mm}^2$$

$$A_{s2} = 2 \times (400 \times 200 + 200 \times 35 + 35 \times 400)$$

$$= 202000 \text{ mm}^2$$

$$M_1 = \frac{112 \times 10^5}{328000} = 34.15 \quad \left(\because M = \frac{V}{A}\right)$$

$$M_2 = \frac{28 \times 10^5}{202000} = 13.86$$

$$\tau_2 = \tau_1 \left(\frac{M_2}{M_1}\right)^2 \quad (\because \tau = M^2)$$

$$= 20 \left(\frac{13.86}{34.15}\right)^2 = 3.29 \text{ min} \approx 3.3 \text{ min}$$

37. Ans: (c)

Sol: Sand casting is used for producing large bells, centrifugal casting is used for producing pulleys, investment casting is used for turbine blades and die casting for producing IC engine pistons and carburetor.

Statement linked Answer Q.No. 38 and 39.

38. Ans: (c)

Sol: Given, $d_s = 10 \text{ mm}$, $h = 250 \text{ mm}$,

Casting size = $100 \times 100 \times 100 \text{ mm}$

$$V_{\text{max}} = \sqrt{2gh} = \sqrt{2 \times 9810 \times 250}$$

$$= 2215 \text{ mm/sec}$$

$$A_c = \frac{\pi}{4} d_s^2 = \frac{\pi}{4} 10^2 = 78.54 \text{ mm}^2$$

$$\text{Flow rate} = V_{\text{max}} \times A_c$$

$$= 173966 \text{ mm}^3/\text{sec} = 1.7 \times 10^5 \text{ mm}^3/\text{sec}$$

39. Ans: (b)

$$\text{Sol: Pouring time} = \frac{\text{Volume (in mm}^3\text{)}}{\text{flow rate (in mm}^3\text{/sec)}}$$

$$= \frac{100^3}{1.7 \times 10^5} = 5.75 \text{ sec}$$

40. Ans: (c)

Sol: Net B.F = $V \cdot g(\rho_L - \rho_{\text{core}})$

$$= \frac{\pi}{4} (0.12)^2 \times 0.18 \times 9.81 \times (11300 - 1600)$$

$$= 193.7 \text{ N}$$

41. Ans: (d)

Sol: Size of cylinder = $D = H$

Side of cube = L

$$M = \frac{\text{Volume}}{\text{Surface Area}}$$

$$M_{\text{cyl}} = \frac{\left(\frac{\pi}{4} D^2\right) H}{(\pi D H) + 2\left(\frac{\pi}{4} D^2\right)} = \frac{D}{6}$$

$$M_{\text{cube}} = \frac{L^3}{6L^2} = \frac{L}{6}$$

$$\frac{\tau_{\text{cyl}}}{\tau_{\text{cube}}} = \left(\frac{D/6}{L/6}\right)^2 = \left(\frac{D}{L}\right)^2 \quad (\because \tau = M^2)$$

42. Ans: (d)

Sol: Loss of head due to friction depends on velocity which is based on the Darcy-Weisbach equation in fluid mechanics.

$$h_f = \frac{fLV^2}{2gD}$$

43. Ans: (c)

$$\text{Sol: } \frac{\pi}{4} D_c^3 = \frac{4}{3} \pi R_s^3$$

where D_c = diameter of cylinder and

R_s = radius of sphere

$$D_c = R_s \sqrt[3]{\frac{16}{3}} = 1.75 R_s$$

$$\frac{\tau_s}{\tau_c} = \left(\frac{(D/6)_s}{(D/6)_c}\right)^2 = \left(\frac{D_s}{D_c}\right)^2 = \left(\frac{2 \times R_s}{1.75 R_s}\right)^2 = 1.31$$

44. Ans: (b)

$$\text{Sol: } t_1 = \frac{\text{volume}}{A \times \sqrt{2gh}}, \quad t_2 = \frac{\text{volume}}{\frac{A}{2} \times \sqrt{2g \times (4h)}}$$

$$\left(\frac{t_2}{t_1}\right) = \left(\frac{\frac{\text{volume}}{\frac{A}{2} \times \sqrt{2g \times (4h)}}}{\frac{\text{volume}}{A \times \sqrt{2gh}}}\right) = 1$$

45. Ans: 7.7

Sol: Net Buoyancy force = $V \cdot g (\rho_{Al} - \rho_{\text{core}})$

$$= \left(\frac{\pi}{4} D^2 L\right) \times g (\rho_{Al} - \rho_{\text{core}})$$

$$= \frac{\pi}{4} 100^2 \times 100 \times 10^{-9} \times 9.81 \times (2600 - 1600)$$

$$= 7.7 \text{ N}$$

46. Ans: (a)

Sol: The given casting is a top riser, hence for top riser the optimum relationship between diameter and height of the riser is $H = D/2$ or $D = 2H$.

$$V = \frac{\pi}{4} D^2 H$$

$$H = \frac{4V}{\pi D^2} \dots \dots (1)$$

Surface area of riser exposed for heat transfer

$$A_s = \pi D H + \frac{\pi}{4} D^2 = \pi D \times \frac{4V}{\pi D^2} + \frac{\pi}{4} D^2 = \frac{4V}{D} + \frac{\pi}{4} D^2$$

For minimum A_s

$$\frac{d(A_s)}{dD} = 0$$

$$\frac{-4V}{D^2} + \frac{\pi}{4} \times 2D = 0;$$

$$\frac{\pi}{2} D = \frac{4V}{D^2} \Rightarrow \frac{\pi}{2} D = \frac{4}{D^2} \times \frac{\pi}{4} \times D^2 H$$

$$\frac{D}{2} = H \Rightarrow D = 2H \Rightarrow \frac{H}{D} = \frac{1}{2} = 1:2$$

47. Ans: 3

Sol: Given,

$$M_r = \frac{V}{A_s} = \frac{D}{6} = \frac{6}{6} = 1 \quad (\because D = H)$$

$$M_c = \frac{7 \times 10 \times 2}{2(7 \times 10 + 10 \times 2 + 2 \times 7)} = \frac{140}{208}$$

$$\frac{\tau_r}{\tau_c} = \left(\frac{M_r}{M_c}\right)^2 = \left(\frac{1}{140/208}\right)^2$$

$$\tau_r = 1.36 \left(\frac{208}{140}\right)^2 = 3.00 \text{ min}$$

48. Ans: (b)

Sol: Given, Volume of riser 1 = Volume of riser 2,

$$\text{i.e., } V_1 = V_2$$

$$\tau_2 = 4\tau_1 \Rightarrow M_2^2 = 4M_1^2$$

$$\left(\frac{V_2}{A_{s2}}\right)^2 = 4 \left(\frac{V_1}{A_{s1}}\right)^2$$

$$\left(\frac{A_{s2}}{A_{s1}}\right)^2 = \left(\frac{1}{4}\right) \Rightarrow \left(\frac{A_{s2}}{A_{s1}}\right) = \left(\frac{1}{4}\right)^{0.5}$$

$$\Rightarrow A_{s2} = \frac{1}{2} A_{s1}$$

49. Ans: (b)

Sol: Given, $\tau \propto \left(\frac{V}{A}\right)^2$

Casting 1 = 6 faces exposed to heat transfer
Casting 2 = 5 faces exposed to heat transfer and 1 face is insulated

$$V_1 = V_2$$

$$A_1 = 6x^2$$

$$A_2 = 5x^2$$

$$\frac{\tau_1}{\tau_2} = \left(\frac{V_2}{A_2} \times \frac{A_1}{V_1}\right)^2 = \left(\frac{A_1}{A_2}\right)^2 = \left(\frac{6x^2}{5x^2}\right)^2 = \left(\frac{6}{5}\right)^2 = \frac{36}{25}$$

50. Ans: 6.16

Sol: $V_s = V_c$

$$\frac{4}{3}\pi R^3 = x^3$$

$$x = R \sqrt[3]{\frac{4}{3}\pi} = 1.612 R$$

$$\frac{\tau_s}{\tau_c} = \left(\frac{(V/A)_s}{(V/A)_c}\right)^2 = \left(\frac{A_c}{A_s}\right)^2$$

$$= \left(\frac{6x^2}{4\pi R^2}\right)^2 = \left(\frac{6 \times (1.62)^2 \times R^2}{4\pi R^2}\right)^2$$

$$\frac{\tau_s}{\tau_c} = 1.54$$

$$\tau_s = 1.54 \times 4 = 6.16 \text{ sec}$$

51. Ans: 10.5 to 10.7

Sol: Side riser = D = H

$$\text{Casting size} = 25 \times 15 \times 5 = 1875 \text{ cm}^3$$

$$\text{Shape factor} = \frac{L+W}{T} = \frac{25+15}{5} = \frac{40}{5}$$

From table corresponding to S.F = 8

$$\frac{V_r}{V_c} = 0.5$$

$$V_r = 0.5 \times V_c = 0.5 \times 1875 = 937.5 \text{ cc}$$

$$= \frac{\pi}{4} D^2 H = \frac{\pi}{4} D^3 = 937.5$$

$$D = H = 10.6 \text{ cm}$$

52. Ans: 0.5625

Sol: $\frac{\tau_{\text{cylindrical}}}{\tau_{\text{cube}}} = ?$

Given that, $H = R \rightarrow \text{Cylinder}$

$x = \text{side of cube} = 2 \times H = 2R = D$

$$V_{\text{cyl}} = \pi R^2 H = \pi R^3$$

$$V_{\text{cube}} = x^3 = D^3 = 8R^3$$

$$A_{\text{cyl}} = 2\pi R H + 2\pi R^2 = 2\pi R^2 + 2\pi R^2 = 4\pi R^2$$

$$A_{\text{cube}} = 6x^2 = 6 \times (2R)^2 = 6 \times 4R^2 = 24R^2$$

$$\frac{\tau_{\text{cyl}}}{\tau_{\text{cube}}} = \left(\frac{V_{\text{cyl}}}{V_{\text{cube}}} \times \frac{A_{\text{cube}}}{A_{\text{cyl}}}\right)^2$$

$$= \left(\frac{\pi R^3}{8R^3} \times \frac{24R^2}{4\pi R^2}\right)^2 = \left(\frac{6}{8}\right)^2 = 0.5625$$

53. Ans: 185 to 195

Sol: Given, Total height (h_t) = 300 mm = 30 cm,

$$d_r = 10 \text{ mm} = 1 \text{ cm};$$

$$g = 9.81 \text{ m/s}^2 = 981 \text{ cm/s}^2$$

Flow rate = velocity of molten metal $\times$ cross sectional area

$$= \sqrt{2 \cdot g \cdot h_t} \times \frac{\pi}{4} (d_r)^2$$

$$= \sqrt{2 \times 981 \times 30} \times \frac{\pi}{4} (1)^2$$

$$= 190.54 \text{ cc/s}$$

54. Ans: 50 (range 49.9 to 50.1)

Sol: $m = 2 \text{ kg}$, $Q = 10 \text{ kW}$

$$\begin{aligned} \text{Time taken for removing latent heat} &= 20 - 10 \\ &= 10 \text{ sec} \end{aligned}$$

$$\text{Time} = \frac{\text{Latent heat}}{Q}$$

$$\text{Latent heat} = \text{time} \times Q = 10 \times 10 = 100 \text{ kJ}$$

$$\text{Latent heat/kg} = \frac{100}{2} = 50 \text{ kJ/kg}$$

55. Ans: (a)

Sol: According to modulus method

$$M_R = 1.2 M_C$$

$$\left(\frac{V}{A_s}\right)_R = 1.2 \left(\frac{V}{A_s}\right)_C$$

Diameter of riser = Height of riser
(for top riser, $D = H$)

$$\Rightarrow D = 6 M_c$$

$$D = 6 \times \frac{\frac{\pi}{4} \times (200)^2 \times 100}{2 \times \frac{\pi}{4} \times 200^2 + \pi \times 200 \times 100}$$

$$D = H = \frac{6 \times 200 \times 100}{400 + 400} = 150 \text{ mm}$$

56. Ans: 1.0927 (range 1.08 to 1.10)

Sol: Ratio of volume of pattern to casting

$$= \frac{1.03^3 \times 100 \times 50 \times 10}{100 \times 50 \times 10} = 1.0927$$

Volume of cavity is assumed to be small and negligible.

57. Ans: 0.9016

Sol: $L_p = L_Q = 500 \text{ mm}$,

Block "P" is rectangular = $100 \text{ mm} \times 200 \text{ mm}$

Block "Q" is square

$$(A_c)_P = (A_c)_Q$$

$$100 \times 200 = x^2$$

$$x = \sqrt{100 \times 200} = 141.42 \text{ mm}$$

$$M_P = \frac{V}{A_s} = \frac{100 \times 200 \times 500}{2[(100 \times 200) + (200 \times 500) + (500 \times 100)]} = 29.411$$

$$M_Q = \frac{V}{A_s} = \frac{(141.42)^2 \times 500}{4 \times 500 \times 141.42 + 2 \times (141.4)^2} = 30.974$$

$$\frac{\tau_P}{\tau_Q} = \left(\frac{M_P}{M_Q} \right)^2 = \left(\frac{29.411}{30.974} \right)^2 = 0.9016$$

58. Ans: 0.81

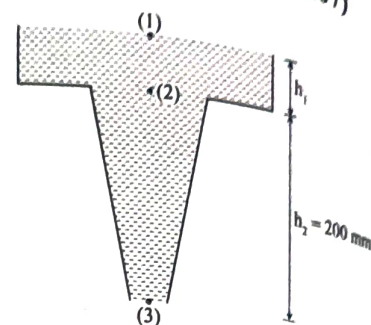
Sol: Given, $l = 300 \text{ mm}$, $\sigma_{\max} = 270 \text{ MPa}$, $E = 100 \text{ GPa}$

$$\sigma_{\max} = E \frac{\Delta l}{l} \Rightarrow 270 = 100 \times 10^3 \times \frac{\Delta l}{300}$$

$$\Delta l = 810 \times 10^{-3} \text{ m} = 0.81 \text{ mm}$$

59. Ans: 2.042 (Range: 2.041 to 2.07)

Sol:



$$d_2 = 20 \text{ mm},$$

$$V_2 = 0.5 \text{ m/s},$$

$$V_2 = \sqrt{2gh_1}$$

$$h_1 = \frac{V_2^2}{2g} = \frac{(0.5)^2}{2 \times 9.8} \times 1000 = 12.755 \text{ mm}$$

$$h_t = h_2 + h_1 = 200 + 12.755 = 212.755$$

$$V_3 = \sqrt{2gh_t} = \sqrt{2 \times 9.81 \times 1000 \times 212.755} = 2042 \text{ mm/sec} = 2.042 \text{ m/sec}$$

60. Ans: (d)

Sol: Given that $h_t = 1500 + 100 = 1600 \text{ mm}$,
 $h_2 = 1500 \text{ mm}$

To avoid aspiration effect

$$A_1 / A_2 = [(h_t - h_2) / h_t]^{0.5} = [(1600 - 1500) / 1600]^{0.5} = (d_1 / d_2)^2$$

$$(d_1 / d_2) = 1/2$$

61. Ans: 51.84 (Range: 42 to 52)

$$\text{Sol: } \frac{\tau_R}{\tau_C} = \left(\frac{m_R}{m_C} \right)^2$$

$$m_C = \frac{80 \times 120 \times 20}{2[(80 \times 120) + (120 \times 20) + (80 \times 20)]}$$

$$m_C = 7.05$$

$$m_R = \frac{d}{6} [\because \text{side riser given}]$$

$$\Rightarrow \frac{m_R}{m_C} = \sqrt{1.5}$$

$$\Rightarrow d = 51.84 \text{ mm}$$